LESSON 01 / 06

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/geometry-via-complex extension (IOQM/RMO core weapon)

Encoding a figure's vertices as complex numbers and proving theorems by algebra is standard olympiad technique (and an occasional JEE Advanced strand). Everything here is derived from Part I (rotation $= \times e^{i\theta}$, ω-identities). The equilateral condition appears in JEE-Advanced-flavoured problems regularly; Ptolemy/Napoleon/van Aubel (next sheets) are classical olympiad theorems. No claim that these "can never appear" in JEE — they are simply at the harder, technique-heavy end.

The engine: rotation is multiplication

KEY RESULT

Rotation by θ about A

$$X' = A + e^{i\theta}(X - A)$$

One line: $X - A$ is the vector $A \to X$; multiplying by $e^{i\theta}$ keeps its length and adds θ to its direction; adding A re-anchors. Quarter turns: $\times i$; half turn: $\times(-1)$; 60°: $\times e^{i\pi/3} = \times(1+\omega)$ (identity 4 of sheet 05) — and $e^{-i\pi/3} = -\omega = \overline{1+\omega}$.

KEY RESULT

Equilateral-triangle condition (orientation-sensitive)

(a, b, c) is equilateral **counterclockwise** $\Leftrightarrow$

$$a + \omega b + \omega^2 c = 0$$

(clockwise ordering gives $a + \omega^2 b + \omega c = 0$). Both directions are one-line proofs below; the condition is the single most-used complex-geometry fact in olympiads.

SHORT DERIVATION

Proof both directions

($\Rightarrow$) CCW equilateral: $c - a = e^{i\pi/3}(b - a) = (1+\omega)(b - a)$ (rotate side ab by $+60^\circ$). Solve: $c = -\omega a + (1+\omega)b$. Substitute:
$a + \omega b + \omega^2 c = a + \omega b - \omega^3 a + (\omega^2 + \omega^3)b = a + \omega b - a - \omega b = 0$ using $\omega^3 = 1$, $\omega^2 + 1 = -\omega$. ∎

($\Leftarrow$) From $a + \omega b + \omega^2 c = 0$: $c = -\omega a - \omega^2 b$ (using $\omega^{-2} = \omega$), so
$c - a = -(1+\omega)a - \omega^2 b = (1+\omega)(b - a)$ (expand and use $1 + \omega + \omega^2 = 0$). Hence $|c - a| = |b - a|$ with a 60° turn: equilateral, CCW. ∎

Check on $a = 0$, $b = 1$, $c = e^{i\pi/3}$: $\omega + \omega^2(1+\omega) = \omega + \omega^2 + \omega^3 = 0$ ✓.

STANDARD APPLICATION

Standard tests (translation dictionary)

Collinear z_1, z_2, z_3: $\frac{z_1 - z_2}{z_1 - z_3} \in \mathbb{R}$ (the ratio of two side-vectors is real). Perpendicular at z_1: that ratio is purely imaginary ($\in i\mathbb{R}$). Equal lengths: $|z_1 - z_2| = |z_3 - z_4|$. Section formula: point dividing $z_1 z_2$ in ratio $m : n$ is $\frac{nz_1 + mz_2}{m+n}$. Angle θ at z between a, b: $\arg \frac{b-z}{a-z} = \pm\theta$ — the sheet-09 arc language. Concyclic: cross-ratio $\frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} \in \mathbb{R}$ (olympiad-level; use only when needed).

KEY RESULT

Area and circumcenter — two more one-liners

$$\text{Area}(\triangle z_1 z_2 z_3) = \tfrac{1}{2}\left|\operatorname{Im}\left((z_2 - z_1)\overline{(z_3 - z_1)}\right)\right|$$

Reason: $(z_2 - z_1)\overline{(z_3 - z_1)}$ is the cross product of the two sides in complex clothing — modulus |product of lengths|, argument = signed angle between them; its imaginary part is the signed twice-area. Shoelace is this formula in coordinates.

Cumcenter u: solve $|u - a|^2 = |u - b|^2 = |u - c|^2$: expanding, $u(\bar{b} - \bar{a}) + \bar{u}(b - a) = |b|^2 - |a|^2$ — one linear equation in $u, \bar{u}$ per pair; two of them, 2×2 solve. No perpendicular bisectors drawn.

WORKED EXAMPLE

Worked — area of $\triangle(1, 3i, -2 + 2i)$; circumcenter of $(1, i, -1)$

$z_2 - z_1 = -1 + 3i$, $z_3 - z_1 = -3 + 2i$;
$(z_2 - z_1)\overline{(z_3 - z_1)} = (-1 + 3i)(-3 - 2i) = 3 + 2i - 9i + 6 = 9 - 7i$. Area $= \frac{1}{2}|-7| = \frac{7}{2}$. (Shoelace check: $\frac{1}{2}|1 \cdot (3 - 2) + 0 + (-2)(0 - 3)| = \frac{1}{2} \cdot 7$ ✓.)

Circumcenter of $1, i, -1$: all on $|z| = 1$, so $u = 0$ (uniqueness of the circumcenter of a non-degenerate triangle). The linear solve confirms: $u(-i - 1) + \bar{u}(i - 1) = |i|^2 - |1|^2 = 0$, satisfied by $u = 0$ ✓.

COMMON TRAP

Traps — rotation toolkit

- Orientation: $a + \omega b + \omega^2 c = 0$ is **CCW-only**; a problem's clockwise labelling needs $\omega \leftrightarrow \omega^2$. If ordering is unstated, check both or phrase conclusions orientation-free.
- $\frac{z_1 - z_2}{z_1 - z_3} \in \mathbb{R}$ excludes $z_1 = z_3$ (division) — degenerate configurations need separate handling.
- Area formula has a $\frac{1}{2}$ and an absolute value; the signed imaginary part tracks orientation.
- Circumcenter 2×2 solve fails when $a = b$ (degenerate); assume distinct vertices.
- $e^{i\pi/3} = 1 + \omega$ and $e^{-i\pi/3} = -\omega$ — mixing these up reverses every rotation in sight.

LESSON 02 / 06

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (classical geometry ↔ algebra)

Ptolemy's theorem/inequality and distance theorems (British Flag and friends) are proved here as pure algebra in $\mathbb{C}$. The identity behind Ptolemy is a 30-second expansion usable in any exam; the equality-condition analysis is olympiad-grade. The JEE-visible content: the identity, the inequality, and the distance expansions — all derivable from Part I.

The Ptolemy identity — no geometry anywhere

KEY RESULT

Identity (valid for ALL four complex numbers a, b, c, d)

$$(a-c)(b-d) = (a-b)(c-d) + (b-c)(a-d)$$

Verify yourself: RHS
$= (ac - ad - bc + bd) + (ab - bd - ca + cd) = ab - ad - bc + cd = (a-c)(b-d)$
✓. (Expansion only — no circle, no ordering, nothing.)

SHORT DERIVATION

Ptolemy's inequality and theorem — one modulus step each

Take moduli and use the triangle inequality:

$$|a-c|\,|b-d|$$

$$|(a-b)(c-d) + (b-c)(a-d)|$$

$$\le |a-b|\,|c-d| + |b-c|\,|a-d|$$

— the product of the diagonals is at most the sum of products of opposite sides, for **any** four points. Equality holds iff $(a-b)(c-d)$ and $(b-c)(a-d)$ point the same direction, i.e. $\frac{(b-c)(a-d)}{(a-b)(c-d)} \in \mathbb{R}_{>0}$; taking arguments this says the angles the chord bd subtends at a and at c are **supplementary** — the points are concyclic in the order a, b, c, d (inscribed-angle theorem, oly-01's collinearity language). That is Ptolemy's theorem: for a cyclic quadrilateral, equality — the diagonal-product formula from school geometry. ∎

Unit-square check ($a = 1, b = i, c = -1, d = -i$): diagonals $2 \times 2 = 4$; sides $\sqrt{2}$: $\sqrt{2}\cdot\sqrt{2} + \sqrt{2}\cdot\sqrt{2} = 4$ ✓.

STANDARD APPLICATION

How to use Ptolemy in an olympiad

1. Encode the four points; write the identity (or just take the cyclic special case if cyclicity is given).
2. For length-product equalities/inequalities ($PA \cdot QC + \ldots$), match to the four factors.
3. Collinearity-ish corollaries: if one "vertex" collapses onto a side, Ptolemy degenerates to the triangle inequality $|a-b| \le |a-c| + |c-b|$ — recognise the degeneration instead of re-deriving.
4. The converse direction (equality ⇒ cyclic) is available when a problem says "prove $ABCD$ cyclic" via lengths.

Distance theorems as expansions — the British Flag

KEY RESULT

British Flag theorem (rectangle form)

For a rectangle A, B, C, D (diagonal pairs A, C and B, D) and **any** point e in the plane:

$$|e-A|^2 + |e-C|^2 = |e-B|^2 + |e-D|^2$$

SHORT DERIVATION

Proof — expand $|z-a|^2 = |z|^2 - z\bar{a} - \bar{z}a + |a|^2$ and collect

Take the square with vertices $0, 1, 1+i, i$; the statement is $|e|^2 + |e-(1+i)|^2 = |e-1|^2 + |e-i|^2$. Expand each:

$$|e-(1+i)|^2 = |e|^2 - e(1-i) - \bar{e}(1+i) + 2,$$

$$|e-1|^2 = |e|^2 - e - \bar{e} + 1,$$

$$|e-i|^2 = |e|^2 + ie - i\bar{e} + 1$$

$LHS = 2|e|^2 - e + ie - \bar{e} - i\bar{e} + 2$; $RHS = 2|e|^2 - e + ie - \bar{e} - i\bar{e} + 2$ — identical term for term. ∎ (For a general rectangle with vertices $0, w, w+v, v$ the same expansion cancels because the vertex constants pair up: $0 + (w+v) = w + v$.)

The pattern worth internalising: $|z-a|^2$ **is a quadratic in $z, \bar{z}$ with clean coefficients** — every rectangle/square distance theorem is "expand and collect".

WORKED EXAMPLE

Worked — the same expansion proves the parallelogram law and the Leibniz $n=2$ case

$|z+w|^2 + |z-w|^2$: expand

$= (|z|^2 + |w|^2 + z\bar{w} + \bar{z}w) + (|z|^2 + |w|^2 - z\bar{w} - \bar{z}w) = 2|z|^2 + 2|w|^2$ ✓ (sheet 01).

And $|z-a|^2 + |z-b|^2 = 2|z-m|^2 + \frac{1}{2}|a-b|^2$ with $m = \frac{a+b}{2}$: expand both sides, compare (or complete the square). Three "theorems", one algebraic habit.

COMMON TRAP

Traps — Ptolemy and expansions

- The identity is unconditional; the **theorem** (equality) needs cyclic order — never claim $|a-c||b-d| = \dots$ for a generic quadrilateral.
- Ptolemy's factors pair as (diagonal, diagonal) vs (opposite side, opposite side) — the ordering a, b, c, d around the circle is part of the statement.
- British Flag is for **rectangles** (points in the plane, any point e — inside or outside); for general quadrilaterals it fails.
- When expanding $|z-a|^2$, note $\overline{(z-a)} = \bar{z} - \bar{a}$ — conjugating the constant correctly is the usual slip.
- Degenerate Ptolemy (a point on a side) gives the triangle inequality — good consistency check, not a new theorem.

LESSON 03 / 06

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (classical theorems, JEE-adjacent technique)
Van Aubel (squares on the sides of a quadrilateral) and Napoleon (equilateral triangles on the sides of a triangle) are olympiad staples proved here by the ω-machinery of Part I. The technique (center formulas + the equilateral condition) is exactly the JEE-Advanced "geometry via complex numbers" strand; the theorems themselves sit at the IOQM/RMO end. Derivable from Part I + oly-01; nothing assumed beyond them.

Outward-square centers

KEY RESULT

Center formula (counterclockwise quadrilateral, outward square on side pq)

$$m_{pq} = \frac{p+q}{2} - \frac{i}{2}(q-p) = \frac{(1+i)p + (1-i)q}{2}$$

Reason: the center of any square on base pq is the midpoint $\pm$ the half-side rotated by 90°; for a CCW quadrilateral the interior lies to the left of $p \to q$, so the outward direction is $-i(q-p)$'s side. (Check on side $0 \to 1$: center $\frac{1-i}{2}$ — below the real axis ✓.)

SHORT DERIVATION

Van Aubel — opposite centers are equal and perpendicular
For CCW a, b, c, d with outward centers $m_{ab}, m_{bc}, m_{cd}, m_{da}$:

$$2(m_{bc} - m_{da})$$

$$((1+i)b + (1-i)c) - ((1+i)d + (1-i)a)$$

$$2(m_{cd} - m_{ab})$$

$$((1+i)c + (1-i)d) - ((1+i)a + (1-i)b)$$

Multiply the second by i and collect: $i \cdot 2(m_{cd} - m_{ab}) = -2(m_{bc} - m_{da})$, i.e.

$$m_{bc} - m_{da} = -i\,(m_{cd} - m_{ab}).$$

One equation, two facts: $\times(-i)$ is a 90° rotation preserving length — the segments are **equal and perpendicular**. ∎ (Numeric check on $0, 1, 1.5 + 1.2i, 0.3 + 0.9i$: both segments ≈ 2.1645 long, ratio pure imaginary ✓.)

Napoleon's Theorem — equilaterals erected on the sides

KEY RESULT

Outward-equilateral center (CCW triangle a, b, c, third vertex via -60° turn)

Third vertex of the outward equilateral on $a \to b$: $a + e^{-i\pi/3}(b-a) = a - \omega(b-a)$. Its center:

$$m_{ab} = \frac{a + b + (a - \omega(b-a))}{3} = \frac{(2+\omega)a + (1-\omega)b}{3}$$

using $e^{-i\pi/3} = -\omega$ and $1 - \omega^2 = 2 + \omega$.

SHORT DERIVATION

Napoleon — the centers form an equilateral triangle

By oly-01's condition, suffices to show $m_{ab} + \omega m_{bc} + \omega^2 m_{ca} = 0$. Compute $3\times$ the left side and collect the a-coefficient:

$$(2+\omega) + \omega^2(1-\omega) = 2 + \omega + \omega^2 - \omega^3 = 1 + \omega + \omega^2$$

$$0$$

and identically for b and c (each coefficient collapses to $1 + \omega + \omega^2 = 0$). Hence $m_{ab} + \omega m_{bc} + \omega^2 m_{ca} = 0$: the outer Napoleon triangle is equilateral. ∎

Numeric check on $(0,0), (4,0), (1,3)$: centers $(2, -1.1547)$, $(3.366, 2.366)$, $(-0.366, 1.789)$; all three mutual distances $\sqrt{14.261}$ ✓.

STANDARD APPLICATION

Technique takeaway — center formulas are linears in the vertices

Any "centers of figures erected on the sides" problem is: (i) write each center as an affine combination of its two base vertices (fixed complex coefficients depending only on the shape erected); (ii) form the target combination; (iii) the vertex coefficients must vanish (or match) — this *is* the theorem. Works for squares (van Aubel), equilaterals (Napoleon), similar triangles, and regular n-gons. The coefficients come from the rotation factors $e^{i\theta}$ of oly-01.

COMMON TRAP

Traps — outward constructions

- Outward vs inward: the sign in $\frac{p+q}{2} \pm \frac{i}{2}(q-p)$ flips the construction; "outward" is fixed by the polygon's orientation (CCW assumed throughout these sheets). Inward squares give a companion theorem (segments still equal/perpendicular) — same proof, sign flip.
- $\omega = e^{2\pi i/3}$ in the equilateral condition, but the 60° turn of a side is $e^{\pm i\pi/3}$: $e^{-i\pi/3} = -\omega$, $e^{i\pi/3} = 1+\omega$ — do not conflate the ω of the condition with the rotation factor.
- Van Aubel is for **any** quadrilateral (convex or not, with outward defined by consistent orientation); Napoleon for any triangle.
- "Centers" are centroids (of the square/equilateral), not vertices or circumcenters — the coefficients above are centroid-specific ($\frac{1}{3}$ for triangles, $\frac{1}{2}+\frac{i}{2}$-type for squares).

LESSON 04 / 06

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (synthesis problems)

Three advanced-but-derivable strands: the Leibniz centroid distance formula (uses only $|z-a|^2$ expansions), unit-circle configurations where $|z_k|=1$ forces a shape (uses $\bar{z}=1/z$), and the map $w=z+\frac{1}{z}$ (Joukowski) collapsing circles to segments. All appear in IOQM/RMO and in coaching "advanced complex numbers" sets; all are derivable from Part I.

Leibniz — distance sums are minimised at the centroid

KEY RESULT

Centroid identity (points $v_1,\dots,v_n$, centroid $c=\frac{1}{n}\sum v_k$, any p)

$$\sum_{k=1}^{n}|p-v_k|^2 = n\,|p-c|^2 + \sum_{k=1}^{n}|v_k-c|^2$$

SHORT DERIVATION

Proof — expand $p-v_k=(p-c)-(v_k-c)$

$$\sum|p-v_k|^2$$

$$=\sum|(p-c)-(v_k-c)|^2$$

$$=n|p-c|^2+\sum|v_k-c|^2-2\mathrm{Re}\left((p-c)\overline{\sum(v_k-c)}\right)$$

and $\sum(v_k-c)=\sum v_k-nc=0$ kills the cross term. ∎ Reading: the second term is the fixed "inertia" of the configuration; the first is minimised at $p=c$ — **the centroid is the point of minimum total squared distance**. Corollary for a regular n-gon of circumradius R centered at 0 with $|p|=1$: sum $=n\cdot 1+nR^2$ — e.g. hexagon $R=2$, $|P|=1$: $6+24=30$ (independent of the direction of P).

Unit-circle configurations — $\bar{z} = 1/z$ is the entry ticket

STANDARD APPLICATION

Four unit numbers summing to zero form a rectangle

If $|z_k| = 1$ and $z_1 + z_2 + z_3 + z_4 = 0$: conjugate the sum and use $\bar{z}_k = 1/z_k$:

$$\frac{1}{z_1} + \frac{1}{z_2}$$

$$= -\left(\frac{1}{z_3} + \frac{1}{z_4}\right) \Rightarrow (z_1 + z_2)\left(\frac{1}{z_1 z_2} - \frac{1}{z_3 z_4}\right)$$

$$= 0$$

(using $z_3 + z_4 = -(z_1 + z_2)$). Either $z_1 + z_2 = 0$ (antipodal pairs $\Rightarrow \{\pm a, \pm b\}$: a cyclic parallelogram = rectangle), or $z_1 z_2 = z_3 z_4$, which makes $\{z_3, z_4\} = \{-z_1, -z_2\}$ (same shape: $t^2 + st + p = 0$ has roots $-1 \times$ those of $t^2 - st + p = 0$). Square exactly when adjacent vertices are 90° apart ($z_2 = \pm i z_1$). ∎

STANDARD APPLICATION

Three unit numbers summing to zero are 120° apart

$|z_1 + z_2| = |z_3| = 1 \Rightarrow 1 = 2 + 2\mathrm{Re}(z_1 \bar{z}_2) \Rightarrow \mathrm{Re}(z_1 \bar{z}_2) = -\frac{1}{2} \Rightarrow$ mutual angles 120°: an equilateral triangle centered at 0; the six points $\{\pm z_k\}$ step by 60°: a **regular hexagon**. (Check: antipodes land exactly halfway between the 120°-spaced points.)

KEY RESULT

Product identity — distances to the n-th roots

$$\prod_{k=0}^{n-1} |z - \zeta_k| = |z^n - 1|$$

(factor $z^n - 1 = \prod (z - \zeta_k)$, take moduli). E.g. vertices of the regular pentagon on $|z| = 1$, distances from $z = 2$: $\prod |2 - \zeta_k| = |32 - 1| = 31$. On the circle itself ($z = e^{i\theta}$): $\prod_k |e^{i\theta} - \zeta_k| = 2\left|\sin\frac{n\theta}{2}\right|$.

Joukowski — $w = z + 1/z$ collapses circles to segments

On $|z| = 1$: $\frac{1}{z} = \bar{z}$, so $w = z + \bar{z} = 2\cos\theta \in [-2, 2]$: the unit circle maps **two-to-one** onto the real segment $[-2, 2]$ (once each at ± 1). Affine images: $w = z + \frac{1}{z} + i$ maps the circle to the segment $[-2 + i, 2 + i]$ — translate the segment by i. Off the circle: $|z| = r \neq 1$ gives the ellipse $\frac{u^2}{(r + 1/r)^2} + \frac{v^2}{(r - 1/r)^2} = 1$ in $w = u + iv$ (derive: $w = re^{i\theta} + r^{-1}e^{-i\theta} = (r + r^{-1})\cos\theta + i(r - r^{-1})\sin\theta$) — circles become ellipses, degenerating to the segment at $r = 1$.

WORKED EXAMPLE

Worked — locus of $w = z + \frac{1}{z} + i$ as $|z| = 1$

$w = 2\cos\theta + i$: the horizontal segment from $-2 + i$ to $2 + i$, traced twice (endpoints once). Every point attained ($\cos\theta = \frac{\operatorname{Re} w}{2}$). For range-type JEE questions ("find the range of $|z + \frac{1}{z} + 2i|$ on $|z| = 1$") the same reduction is sheet-10 method: $2\cos\theta + 2i \Rightarrow [2, 2\sqrt{2}]$ ✓ consistent.

COMMON TRAP

Traps — synthesis

- $\bar{z} = 1/z$ requires $|z| = 1$; conjugating a relation to introduce $1/z$ is illegal off the circle.
- Rectangle conclusion from four-unit-sum-zero assumes all four on the unit circle; drop that and the statement dies (z_k arbitrary with zero sum says nothing).
- Joukowski collapse is **two-to-one** (z and $1/z$ map to the same w); locus answers list the image set, not parametrisations.
- The product identity's index runs 0 to $n - 1$ (includes $z = 1$'s factor $|z - 1|$); the $\prod(1 - \zeta_k) = n$ formula of sheet 06 is its $z = 1$ cousin **excluding** ζ_0 — keep the two consistent (at $z = 1$: $|z^n - 1| = 0$; the nonzero cousin excludes $k = 0$).
- Leibniz needs the **centroid**; using the circumcenter instead gives the wrong "minimum" statement (the circumcenter minimises max distance, a different problem).

LESSON 05 / 06

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad extension (deeper number theory; IOQM/RMO+)

The quadratic Gauss sum $G(n)=\sum_{k=0}^{n-1}\zeta^{k^2}$ is classical olympiad/number-theory material (the analytic heart of quadratic reciprocity). Its **evaluation mechanism** — square, re-index, let the vanishing sum filter — uses nothing but Part I sheets 05–06. The full reciprocity connection is quoted as context, not required. This is specialised extension material, not core JEE; nothing prevents an exam from asking the small cases ($n=3,4,5$) computationally.

The object and its closed form

For $\zeta=e^{2\pi i/n}$ define $G(n)=\sum_{k=0}^{n-1}\zeta^{k^2}$ (the exponent is k^2 — this is why the vanishing-sum filter does not kill it outright).

KEY RESULT

Closed form (phase locked to n mod 4)

$$G(n)=\sqrt{n}\ \text{ if } n\equiv 1\ (4);\qquad =i\sqrt{n}\ \text{ if } n\equiv 3\ (4)$$

$$G(n)=(1+i)\sqrt{n}\ \text{ if } n\equiv 0\ (4);\qquad =0\ \text{ if } n\equiv 2\ (4)$$

Mechanism I: square, then re-index

SHORT DERIVATION

$|G(n)|^2$ — the double sum the filter can collapse

Each term has modulus 1, so $|G(n)|^2=\sum_{k,l}\zeta^{k^2-l^2}$. Put $k=l+h$: $k^2-l^2=2lh+h^2$, and

$$|G(n)|^2=\sum_{h=0}^{n-1}\zeta^{h^2}\sum_{l=0}^{n-1}\zeta^{2lh}$$

- n odd: $n\mid 2h\Rightarrow n\mid h$: only $h=0$ survives: $|G(n)|^2=n$.
- $n=2m$: $n\mid 2h\Leftrightarrow m\mid h$: $h=0$ and $h=m$ survive; $\zeta^{m^2}=e^{\pi i m}=(-1)^m$, so $|G(n)|^2=n(1+(-1)^m)$: 0 for m odd ($n\equiv 2\pmod 4$), $2n$ for m even ($n\equiv 0\pmod 4$).

Squaring is the whole trick: it turns a 1-dimensional sum into a 2-dimensional one the filter collapses. ∎

Mechanism 2: the phase, case by case

Since $k^2 \equiv (n-k)^2 \pmod n$, terms pair: $G(n) = 1 + 2\sum_{k=1}^{(n-1)/2} \zeta^{k^2}$ for odd n.

WORKED EXAMPLE

Small cases, computed exactly

$G(3) = 1 + 2\omega = 1 + 2(-\frac{1}{2} + i\frac{\sqrt{3}}{2}) = i\sqrt{3}$ ✓ ($n = 3 \equiv 3$; phase i).

$G(5) = 1 + 2(\zeta + \zeta^4) = 1 + 4\cos\frac{2\pi}{5} = 1 + (\sqrt{5} - 1) = \sqrt{5}$ ✓ ($n \equiv 1$; phase 1), using $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.

$G(7) = 1 + 2(\zeta + \zeta^2 + \zeta^4) = i\sqrt{7}$ ✓ ($n = 7 \equiv 3$); the exponent residues $k^2 \pmod 7$ run $0, 1, 4, 2, 2, 4, 1$.

$G(8)$: residues $k^2 \pmod 8$ are $0, 1, 4$ with 4 twice and 1 four times:
$G(8) = 2(1 + \zeta^4) + 4\zeta = 4 \cdot \frac{1+i}{\sqrt{2}} = (1+i)\sqrt{8}$ ✓ ($n \equiv 0$).

$G(2)$: here $\zeta = e^{2\pi i/2} = -1$ and $k = 0, 1$ give $\zeta^0 + \zeta^1 = 1 + (-1) = 0$ ✓ (vanishes, as the mod-4 rule predicts). Likewise $G(6) = G(10) = 0$.

STANDARD APPLICATION

How to use this in a problem

1. **Vanishing case first**: if $n \equiv 2 \pmod 4$, the answer is 0 before any arithmetic.
2. Odd n: pair k with $n-k$, evaluate the half-sum in trig ($\cos\frac{2\pi k}{n}$ values) when n is small; or quote the phase rule with the modulus $\sqrt{n}$.
3. The Legendre-symbol form $G(p) = \sum_{k=0}^{p-1} (\frac{k}{p})\zeta^k$ for odd primes p is the same number; $G(p) = \sqrt{p}$ or $i\sqrt{p}$ by $p \pmod 4$ is quadratic reciprocity's analytic core — quote as context.
4. Verification habit: compute $|G(n)|^2$ numerically (should be n, 0, or $2n$ per the table) — one number confirms the phase branch.

COMMON TRAP

Traps — Gauss sums

- $n \equiv 2 \pmod 4$: the sum is exactly 0 — not "small"; the two surviving filter contributions cancel (1 and $(-1)^m$ with m odd).
- The phase: $n \equiv 1$: real positive; $n \equiv 3$: purely imaginary positive; $n \equiv 0$: $(1+i)\sqrt{n}$ (argument $\frac{\pi}{4}$) — mixing up $n \pmod 4$ with $n \pmod 3$ because "odd/even" feels like enough is the standard slip.
- The exponent is k^2 **mod** n inside ζ^{k^2} only implicitly: $\zeta^{k^2} = \zeta^{k^2 \pmod n}$ — when pairing residues, reduce mod n first.
- $G(2) = 0$ uses $\zeta = e^{2\pi i/2} = -1$; reading ζ as $i = e^{2\pi i/4}$ (the $n = 4$ root) gives the wrong sum $1 + i$. ζ is always $e^{2\pi i/n}$ for the n of the problem.
- ζ depends on n; every line must use the same n throughout.

LESSON 06 / 06

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (deeper theory; IOQM/RMO+)

Three structure theorems about polynomials and equilateral triangles. **Gauss–Lucas** (critical points stay in the convex hull of the roots) is the complex substitute for Rolle's theorem; **Marden** pinpoints the critical points of a cubic as ellipse foci; **Pompeiu** says PA, PB, PC from an equilateral triangle always form a triangle. Each proof here is a few lines of Part I algebra; the theorems themselves are beyond the core JEE syllabus sentence and are labelled accordingly. No claim that any of this "can never appear" in JEE Advanced — the computational corollaries (critical points of a cubic; distance-triangle inequalities) are ordinary Part I manipulations.

Gauss–Lucas — critical points cannot leave the hull

KEY RESULT

Theorem

Let $p(z) = c\prod_{k=1}^{n}(z - r_k)$. Then every zero of $p'(z)$ lies in the **convex hull** of $\{r_1, \dots, r_n\}$.

SHORT DERIVATION

Proof — the logarithmic derivative is a sum of point charges

Logarithmic differentiation: $\frac{p'(z)}{p(z)} = \sum_{k=1}^{n}\frac{1}{z - r_k}$. If z lies outside the hull, a line separates it from all roots: for some real θ, $\operatorname{Re}\left(e^{-i\theta}(z - r_k)\right) > 0$ for all k. Then

$$\operatorname{Re}\left(e^{i\theta}\frac{p'(z)}{p(z)}\right)$$

$$= \sum_{k=1}^{n}\frac{\operatorname{Re}\left(e^{-i\theta}(z - r_k)\right)}{|z - r_k|^2} > 0$$

so $p'(z)/p(z) \neq 0$, hence $p'(z) \neq 0$ — contradiction. ∎

Corollary (Rolle in complex form): all roots real ⇒ hull is a segment of the real line ⇒ every critical point is real — no intermediate-value argument anywhere.

WORKED EXAMPLE

Worked — critical points of $p(z) = z(z-2)(z-1-2i)$

Expand: $p(z) = z^3 + (-3-2i)z^2 + (2+4i)z$; $p'(z) = 3z^2 + (-6-4i)z + (2+4i)$.
Discriminant: $b^2 - 4ac = (20+48i) - (24+48i) = -4$, so $\sqrt{\text{disc}} = \pm 2i$:

$$z = \frac{6+4i \pm 2i}{6} \in \left\{1+i,\ 1+\tfrac{i}{3}\right\}.$$

Both lie inside the triangle $(0, 2, 1+2i)$ (half-plane check: $y \geq 0$, $y \leq 2x$, $y \leq 4-2x$ ✓), exactly as Gauss-Lucas demands. Shifting the centroid $1+\frac{2}{3}i$ to 0 sends the two points to $\pm\frac{i}{3}$ — symmetric about the centroid, which is Marden's signature.

STANDARD APPLICATION

Marden's theorem (quote-level)

For a cubic whose roots are the vertices of a triangle, the two roots of p' are the **foci of the Steiner inellipse** (the ellipse tangent to the three sides at their midpoints). Gauss-Lucas puts them inside the triangle; Marden identifies them exactly. Test with distances: each side midpoint m_k satisfies $|m_k - f_1| + |m_k - f_2| =$ the same constant. For an equilateral triangle the foci coincide at the center and p' has a double root there.

Pompeiu — three distances that always make a triangle

KEY RESULT

Theorem

ABC equilateral, P any point of the plane: PA, PB, PC are the side lengths of a (possibly degenerate) triangle; degenerate **exactly** when P lies on the circumcircle of ABC.

SHORT DERIVATION

Proof — the equilateral condition cancels P

Take $\omega = e^{2\pi i/3}$ with $A + \omega B + \omega^2 C = 0$ (CCW, oly-01). Put $u = P - A$, $v = P - B$, $w = P - C$; then

$$u + \omega v + \omega^2 w = P(1 + \omega + \omega^2) - (A + \omega B + \omega^2 C)$$

$$0$$

Rotate this one identity to read all three triangle inequalities: as it stands, $u = -\omega v - \omega^2 w \Rightarrow |u| \leq |v| + |w|$; multiplied by ω^2: $|v| \leq |u| + |w|$; by ω: $|w| \leq |u| + |v|$. So PA, PB, PC form a triangle. Equality in $|u| \leq |v| + |w|$ needs $\omega v, \omega^2 w$ parallel with the same sign — precisely P on the arc BC of the circumcircle; by symmetry, on the whole circle. ∎

Concrete: equilateral $1, \omega, \omega^2$ (circumcircle $= |z| = 1$): every $P = e^{i\theta}$ makes $PA + PB = PC$ for the largest distance (degenerate), while $|P| = 2$ gives a genuine triangle (e.g. at $\theta = \frac{\pi}{3}$: $1.414 + 2.504 > 3.120$ ✓).

STANDARD APPLICATION

Why these theorems are the same trick

Gauss–Lucas: expand $\frac{p'}{p}$ and take real parts. Pompeiu: expand the ω-relation and take moduli. Ptolemy (oly-02): expand an identity and take moduli. **Expand, then extract a geometric statement from an algebraic equality/inequality** — the complex-bash method in three costumes. When an olympiad problem smells like "points, distances, and a hidden regular polygon", this is the family to try first.

COMMON TRAP

Traps — structure theorems

- Gauss–Lucas bounds by the convex **hull**, not the circumcircle ("zeros of p' lie in the circle through the roots" is a weaker, half-remembered statement that is sometimes false).
- The hull can be a segment (collinear roots) or the full triangle; boundary cases (critical point *on* the hull edge) are allowed by the theorem.
- Pompeiu degeneracy is *exactly* on the circumcircle — inside the triangle the distance-triangle is strictly non-degenerate (so "PA, PB, PC can never form a triangle" is wrong for interior points).
- Pompeiu needs **equilateral**; for general triangles the three distances need not satisfy any clean inequality beyond the obvious pairwise ones.
- The orientation in $A + \omega B + \omega^2 C = 0$ is CCW; swapping B, C swaps $\omega \leftrightarrow \omega^2$ and the Pompeiu rotation bookkeeping follows suit.